

2024, Mar. 2024 Release LOC:Acute Adult

Cystic Fibrosis

Utilization Benchmarks: *Multiple options available, see table below for options*

Overview

Select Day

Initial review ⁽¹⁾

Episode Day 1

Episode Day 2-5

Episode Day 6

Episode Day 7-9

Episode Day 10

Discharge Screens ⁽⁶⁶⁾

Utilization Benchmarks

Notes

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Overview

Introduction:

Cystic fibrosis is a multisystem disorder caused by a genetic mutation that produces increased sodium levels in combination with ineffective chloride absorption creating thickened mucus. Complications of mucus production include the obstruction of the pancreatic ducts, sinus tracts, and airways. Antibiotic therapy usually consists of a 10-14 day duration and is based on response to treatment. Severe exacerbations or prolonged symptom recovery may require additional treatment up to 21 days. Exacerbations that are treated appropriately prevent further damage to the lungs and have better survival rates. The rate of decline in lung function is associated with the frequency and severity of pulmonary exacerbations, defined as the acute onset of respiratory impairment. The etiology of a pulmonary exacerbation is multifactorial and may include: decreased immunity, changes in airway microbiology (e.g., invasion by new organisms or strains of existing organisms, increased concentration of colonizing flora, or growth of existing strains), inflammatory response, and poor compliance with prescribed maintenance therapy. While outpatient management of pulmonary exacerbations may be an option for select patients, hospitalization is recommended when outpatient resources cannot match the close observation and intensity of therapy required in the inpatient setting.

Evaluation/Treatment:

Currently, no accepted criteria exist to confirm a pulmonary exacerbation. Diagnosis is based on a constellation of clinical signs and symptoms including: changes in cough, sputum production, work of breathing, energy level, appetite, weight loss, and pulmonary function tests. Treatment consists of intravenous anti-infective therapy and intensification of pulmonary clearance measures. Anti-infective therapy should be based on prior culture results, with modification determined by new culture results and clinical response. Chronic infection is the persistent isolation of an organism on repeat cultures over 6 months to a year leading to a high rate of acquired antimicrobial resistance. In addition to anti-infective treatment, guidelines recommend continuation of baseline medication regimens and intensification of airway clearance therapy in the overall management of a pulmonary exacerbation. The treatment regimen should also address the common comorbidities associated with cystic fibrosis:

- Malnutrition: Increased energy expenditure during a pulmonary exacerbation requires optimization of nutrition via pancreatic enzyme replacement therapy and a high calorie, high protein diet.
- Cystic Fibrosis-Related Diabetes: Insulin requirements may increase during a pulmonary exacerbation.
- Constipation: Adherence to a strict bowel regimen in patients with impaired gastrointestinal motility is necessary to prevent obstruction.
- Renal disease: Chronic use of nephrotoxic antibiotics (e.g., aminoglycosides) contributes to long-term renal damage. Close monitoring of renal function is recommended in patients receiving intravenous

aminoglycosides.

Reference: (Talwalkar and Murray, Clin Chest Med 2016, 37: 69-81; Gillespie et al., Int J Hepatol, 2017: 5128760; Bos et al., J Cyst Fibros 2017, 16: 13-23; Balaguer and Gonzalez de Dios, Cochrane database of systematic reviews 2012, 3: CD001917)

InterQual® criteria are derived from the systematic, continuous review and critical appraisal of the most current evidence-based literature and include input from our independent panel of clinical experts. To generate the most appropriate recommendations, a comprehensive literature review of the clinical evidence was conducted. Sources searched included PubMed, Agency for Healthcare Research and Quality (AHRQ) Comparative Effectiveness Reviews, the National Institute for Health and Care Excellence (NICE), the Centers for Medicare and Medicaid Services (CMS), The Cochrane Library, and The Joint Commission. Other medical literature databases, medical content providers, data sources, regulatory body websites, and specialty society resources may also have been utilized. Relevant studies were assessed for risk of bias following principles described in the *Cochrane Handbook*. The resulting evidence was assessed for consistency, directness, precision, effect size, and publication bias. Observational trials were also evaluated for the presence of a dose-response gradient and the likely effect of plausible confounders. The content is based on a variety of references which are cited at specific criteria points throughout the subset.

In an effort to address inequities in medicine and improve appropriate access to care, multiple medical societies have been reevaluating the use of adjustments made for a patient's race and updating their calculators and reference equations when reporting diagnostic results.

In 2023, the American Thoracic Society issued updated guidance on reporting of pulmonary function tests (PFTs), advocating for a race-neutral average reference equation to determine the baseline (i.e., predicted normal) for a given patient (Bhakta et al., Am J Respir Crit Care Med 2023, 207: 978-995). The reference equation previously included lower predicted normal values for patients of certain races and ethnicities (in the US, for Black and Asian patients); a practice which has not been found to have biological validity and may lead to misclassification of diagnoses and disease severity in racial or ethnic minorities (Vyas et al., N Engl J Med 2020, 383: 874-882).

Since not all spirometer manufacturers and PFT laboratories have switched over to the new race-neutral average reference equation, clinical judgement and consideration of patient-specific factors should be used when interpreting PFT results and deciding whether patients with borderline reported PFT values in the presence of concerning symptoms or signs should have further evaluation or treatment.

(Excludes PO medications unless noted)

Select Day, One:● **Initial review, One:** ⁽¹⁾

(From arrival in ED through decision to admit)

(Symptom or finding within 24h)

● **ACUTE, One:** ⁽²⁾● Pulmonary exacerbation, ≥ **Two:** ^(3, 4)

- Cough and increased intensity or frequency
- Decreased exercise tolerance

● Failed outpatient treatment and, **All:** ⁽⁵⁾

- Anti-infective (new)
- Nebulizer treatment with increased dose or frequency
- No evidence of clinical improvement
- Hemoptysis, increased or new onset ⁽⁶⁾

● Hypoxia and, ≥ **One:**

- New supplemental oxygen requirement
- Increased supplemental oxygen requirement
- O₂ sat decreased by > 4%(0.04) from baseline ⁽⁷⁾

● Increased work of breathing, **Both:**

- Respiratory rate > 30/min, sustained ⁽⁸⁾

● Symptom, ≥ **One:**

- Dyspnea, worsening
- Use of accessory muscles
- Retractions
- New pulmonary infiltrate by chest x-ray (CXR)

● FEV₁, **Both:**

- Decreased by ≥ 10%(0.10) from baseline ^(9, 10)
- Performed within last 3 months ⁽⁹⁾
- Sputum production increased
- Temperature ≥ 100.4°F(38.0°C) ⁽¹¹⁾
- Weight loss ⁽¹²⁾
- Wheezing, new onset

● **INTERMEDIATE, One:** ⁽¹³⁾● Respiratory insufficiency, ≥ **One:**

- Arterial Po₂ 40-54 mmHg(5.3-7.2 kPa)
- Arterial Pco₂ 50-59 mmHg(6.7-7.9 kPa) and pH < 7.35 without mental status change ⁽¹⁴⁾
- Venous Pco₂ 50-59 mmHg(6.7-7.9 kPa) and pH < 7.35 without mental status change ⁽¹⁴⁾
- O₂ sat < 89%(0.89) and < baseline ⁽⁷⁾

● **CRITICAL, ≥ One:** ⁽¹⁵⁾● Hemodynamic instability unresponsive to ≥ 2 liter fluid bolus, requiring ongoing fluid resuscitation ≤ 24h and, ≥ **One:** ⁽¹⁶⁾

- MAP < 65 mmHg
- Systolic blood pressure < 90 mmHg and < baseline
- Hemodynamic monitoring, invasive ⁽¹⁷⁾
- Hemoptysis and bronchial artery embolization ⁽¹⁸⁾
- HFNC and respiratory monitoring every 1-2h ^(19, 20, 21)
- Mechanical ventilation

● NIPPV and, ≥ **One:** ⁽²²⁾

- Arterial Po₂ ≤ 39 mmHg(5.2 kPa)
- Arterial or venous Pco₂ ≥ 60 mmHg(8.0 kPa) and, **One:**
 - COPD and arterial or venous pH ≤ 7.24
 - No hx of COPD
- Vasopressor, vasodilator, or inotrope, continuous (excludes low or fixed dose for end stage heart failure)

● **Episode Day 1, One:**

(Symptom or finding within 24h)

● **ACUTE, One:** ⁽²⁾● Pulmonary exacerbation, **Both:** ^(3, 4)● Finding, **≥ Two:**

- Cough and increased intensity or frequency
- Decreased exercise tolerance

● Failed outpatient treatment and, **All:** ⁽⁵⁾

- Anti-infective (new)
- Nebulizer treatment with increased dose or frequency
- No evidence of clinical improvement
- Hemoptysis, increased or new onset ⁽⁶⁾

● Hypoxia and, **≥ One:**

- New supplemental oxygen requirement
- Increased supplemental oxygen requirement
- O₂ sat decreased by > 4%(0.04) from baseline ⁽⁷⁾

● Increased work of breathing, **Both:**

- Respiratory rate > 30/min, sustained ⁽⁸⁾

● Symptom, **≥ One:**

- Dyspnea, worsening
- Use of accessory muscles
- Retractions
- New pulmonary infiltrate by chest x-ray (CXR)

● FEV₁, **Both:**

- Decreased by ≥ 10%(0.10) from baseline ^(9, 10)
- Performed within last 3 months ⁽⁹⁾
- Sputum production increased
- Temperature ≥ 100.4°F(38.0°C) ⁽¹¹⁾
- Weight loss ⁽¹²⁾
- Wheezing, new onset

● Intervention, **Both:** ⁽²³⁾● Airway clearance therapy, **One:** ⁽²⁴⁾

- At least 3x/24h
- Contraindicated due to hemoptysis ⁽²⁵⁾
- Anti-infective

● **INTERMEDIATE, Both:** ⁽¹³⁾● Respiratory insufficiency **≥ One:**

- Arterial Po₂ 40-54 mmHg(5.3-7.2 kPa)
- Arterial Pco₂ 50-59 mmHg(6.7-7.9 kPa) and pH < 7.35 without mental status change ⁽¹⁴⁾
- Venous Pco₂ 50-59 mmHg(6.7-7.9 kPa) and pH < 7.35 without mental status change ⁽¹⁴⁾

● O₂ sat, **One:**

- O₂ sat < 89%(0.89) and < baseline ⁽⁷⁾
- Room air assessment not clinically appropriate ⁽²⁶⁾

● Intervention, **Both:** ^(4, 12)

- Anti-infective

● Oxygen, **One:**● Respiratory intervention, **Both:** ^(24, 27)● Intervention, **≥ One:**

- HFNC ⁽²⁰⁾
- Oxygen ≥ 40%(0.40) ⁽¹⁹⁾
- Respiratory interventions every 3-4h ^(24, 27)
- NIPPV ⁽²²⁾

● **CRITICAL, ≥ One:** ⁽¹⁵⁾

- Hemodynamic instability unresponsive to ≥ 2 liter fluid bolus, requiring ongoing fluid resuscitation ≤ 24 h and, $\geq$ **One:** ⁽¹⁶⁾
 - MAP < 65 mmHg
 - Systolic blood pressure < 90 mmHg and $<$ baseline
- Hemodynamic monitoring, invasive ⁽¹⁷⁾
- Hemoptysis and bronchial artery embolization ^(18, 28)
- HFNC and respiratory monitoring every 1-2h ^(19, 20, 21)
- Mechanical ventilation
- NIPPV and, $\geq$ **One:** ⁽²²⁾
 - Arterial $\text{Po}_2 \leq 39$ mmHg(5.2 kPa)
 - Arterial or venous $\text{Pco}_2 \geq 60$ mmHg(8.0 kPa) and, **One:**
 - COPD and arterial or venous pH ≤ 7.24
 - No hx of COPD
 - Vasopressor, vasodilator, or inotrope, continuous (excludes low or fixed dose for end stage heart failure)
- **Episode Day 2-5, One:**
 - **ACUTE, Both:**
 - Airway clearance therapy, **One:** ⁽²⁴⁾
 - At least 3x/24h
 - Contraindicated due to hemoptysis ⁽²⁵⁾
 - Anti-infective
 - **INTERMEDIATE, $\geq$ One:**
 - Anti-infective desensitization ≤ 24 h ⁽²⁹⁾
 - HFNC and respiratory monitoring every 3-4h ^(20, 21)
 - NIPPV ⁽²²⁾
 - Oxygen $\geq 40\%(0.40)$, ≤ 2 d ⁽¹⁹⁾
 - **CRITICAL, $\geq$ One:**
 - AKI and, **Both:** ⁽³⁰⁾
 - Finding, $\geq$ **One:**
 - Potassium > 6.0 mEq/L(6.0 mmol/L) and, $\geq$ **One:**
 - Loss of P wave
 - Multifocal PVCs
 - Neuromuscular deficit ⁽³¹⁾
 - Widening QRS
 - Sodium, **One:**
 - < 120 mEq/L(120 mmol/L) and volume overload
 - > 160 mEq/L(160 mmol/L)
 - Intervention, $\geq$ **One:**
 - Calcium chloride
 - Calcium gluconate
 - Glucose 50%(0.50) with insulin ⁽³²⁾
 - Diuretic, continuous
 - Dialysis, **One:**
 - Hemodialysis ≤ 7 d since initiation
 - Peritoneal dialysis ≤ 7 d since initiation
 - IV fluid resuscitation ≤ 2 d ⁽³³⁾
 - CRRT ⁽³⁴⁾
 - DIC and blood product transfusion ⁽³⁵⁾
 - ECMO or ECLS ⁽³⁶⁾
 - Hemodynamic instability unresponsive to ≥ 2 liter fluid bolus, requiring ongoing fluid resuscitation ≤ 24 h and, $\geq$ **One:** ⁽¹⁶⁾
 - MAP < 65 mmHg
 - Systolic blood pressure < 90 mmHg and $<$ baseline
 - Hemodynamic monitoring, invasive ⁽³⁷⁾

- Hemoptysis > 240ml/24h and, ≥ **One:** ⁽¹⁸⁾
 - Lung resection required (**use General Surgical subset**) ⁽³⁸⁾
 - Blood product transfusion
 - Bronchial artery embolization performed within last 24h ⁽²⁸⁾
 - Post bronchial artery embolization monitoring ≤ 24h
- HFNC and respiratory monitoring every 1-2h ^(19, 20, 21)
- Mechanical ventilation
- NIPPV and, ≥ **One:** ⁽²²⁾
 - FiO₂ > 50%(0.50)
 - Respiratory intervention every 1-2h ⁽²⁷⁾
- Vasopressor, vasodilator, or inotrope, continuous (excludes low or fixed dose for end stage heart failure)
- **Episode Day 6, One:**
 - **ACUTE, One:**
 - **Early Responder**, discharge expected today if clinically stable last 12h, **All:** ⁽³⁹⁾
 - Anti-infective regimen, **One:**
 - Complete
 - Established and tolerated, **One:**
 - Transition to PO anti-infective
 - Vascular access established for IV anti-infective
 - O₂ sat ≥ 92%(0.92) or within acceptable limits ⁽⁴⁰⁾
 - Pulmonary function return to baseline or improving ^(7, 41)
 - Tolerating PO or nutritional route established ^(42, 43)
 - Treatment needs manageable in post acute setting ⁽⁴⁴⁾
 - No complication or active comorbidity relevant to this episode of care ⁽⁴⁵⁾
 - **Partial responder**, not clinically stable for discharge and requires continued stay, ≥ **One:**
 - Inadequate clinical improvement and, **Both:** ⁽⁴¹⁾
 - Finding, ≥ **One:**
 - FEV₁ < 80%(0.80) of baseline ^(9, 10)
 - Hemoptysis, persistent
 - O₂ sat < 92%(0.92) and < baseline requiring supplemental oxygen ^(7, 19)
 - Pulmonary infiltrates worsening by imaging
 - Sputum production increasing
 - Intervention, **Both:**
 - Airway clearance therapy, **One:** ⁽²⁴⁾
 - At least 3x/24h
 - Contraindicated due to hemoptysis ⁽²⁵⁾
 - Anti-infective
 - Anti-infective regimen incomplete and, **Both:**
 - Barrier to discharge, ≥ **One:** ⁽⁴⁴⁾
 - Patient or caregiver unable to adhere to treatment regimen
 - Treatment intensity exceeds resources provided in post acute setting
 - Intervention, **Both:**
 - Airway clearance therapy, **One:** ⁽²⁴⁾
 - At least 3x/24h
 - Contraindicated due to hemoptysis ⁽²⁵⁾
 - Anti-infective
 - AKI, **Both:** ⁽³⁰⁾
 - Finding, ≥ **One:**
 - Oliguria ^(46, 47)
 - Creatinine, **One:**
 - ≥ 1.5x baseline and chronic kidney disease
 - ≥ 1.5x ULN ⁽⁴⁸⁾
 - GFR > 25%(0.25) decrease from baseline ⁽⁷⁾

- Dialysis initiated
- **Intervention, ≥ One:**
 - Volume expander ≤ 2d ^(49, 50)
 - Diuretic ≥ 2x/24h, ≤ 3d
 - Dialysis (initial) ≤ 7d since initiation ^(51, 52)
 - Dialysis discontinued requiring lab monitoring ≤ 2d since discontinuation ⁽⁵³⁾
 - Immunosuppressant therapy ≤ 3d ^(54, 55, 56)
- Adverse anti-infective drug reaction requiring change in medication ≤ 24h ⁽⁵⁷⁾
- Anti-infective level non-therapeutic and adjustment in dose or frequency ≤ 2d
- **Blood glucose ≤ 24h, All:** ⁽⁵⁸⁾
 - **Finding, ≥ One:**
 - > 250 mg/dL(13.9 mmol/L)
 - < 70 mg/dL(3.9 mmol/L)
 - Blood glucose monitoring
 - Insulin adjustment or dose held ^(58, 59)
- *C. difficile* infection, actual or suspected and, **Both:** ⁽⁶⁰⁾
 - Anti-infective (includes PO) ≤ 5d ^(61, 62)
 - Inadequate oral intake ⁽⁶³⁾
- **Chest tube and, ≥ One:** ⁽⁶⁴⁾
 - Suction, continuous
 - Drainage ≥ 250 mL/24h
 - Repositioning within last 24h
 - Water seal within last 24h
- Continued weight loss and initiation or advancement of tube feeding ≤ 3d
- Post Critical care ≤ 24h
- **INTERMEDIATE, ≥ One:**
 - Anti-infective desensitization ≤ 24h ⁽²⁹⁾
 - HFNC and respiratory monitoring every 3-4h ^(20, 21)
 - NIPPV ⁽²²⁾
 - Oxygen ≥ 40%(0.40), ≤ 2d ⁽¹⁹⁾
- **CRITICAL, ≥ One:**
 - **AKI and, Both:** ⁽³⁰⁾
 - **Finding, ≥ One:**
 - **Potassium > 6.0 mEq/L(6.0 mmol/L) and, ≥ One:**
 - Loss of P wave
 - Multifocal PVCs
 - Neuromuscular deficit ⁽³¹⁾
 - Widening QRS
 - **Sodium, One:**
 - < 120 mEq/L(120 mmol/L) and volume overload
 - > 160 mEq/L(160 mmol/L)
 - **Intervention, ≥ One:**
 - Calcium chloride
 - Calcium gluconate
 - Glucose 50%(0.50) with insulin ⁽³²⁾
 - Diuretic, continuous
 - **Dialysis, One:**
 - Hemodialysis ≤ 7d since initiation
 - Peritoneal dialysis ≤ 7d since initiation
 - IV fluid resuscitation ≤ 2d ⁽³³⁾
 - CRRT ⁽³⁴⁾
 - DIC and blood product transfusion ⁽³⁵⁾
 - ECMO or ECLS ⁽³⁶⁾

- Hemodynamic instability unresponsive to ≥ 2 liter fluid bolus, requiring ongoing fluid resuscitation ≤ 24 h and, $\geq$ **One:** ⁽¹⁶⁾
 - MAP < 65 mmHg
 - Systolic blood pressure < 90 mmHg and $<$ baseline
- Hemodynamic monitoring, invasive ⁽¹⁷⁾
- Hemoptysis > 240 ml/24h and, $\geq$ **One:** ⁽¹⁸⁾
 - Lung resection required (**use General Surgical subset**) ⁽³⁸⁾
 - Blood product transfusion
 - Bronchial artery embolization performed within last 24h ⁽²⁸⁾
 - Post bronchial artery embolization monitoring ≤ 24 h
- HFNC and respiratory monitoring every 1-2h ^(19, 20, 21)
- Mechanical ventilation
- NIPPV and, $\geq$ **One:** ⁽²²⁾
 - FiO₂ $> 50\%$ (0.50)
 - Respiratory intervention every 1-2h ⁽²⁷⁾
- Vasopressor, vasodilator, or inotrope, continuous (excludes low or fixed dose for end stage heart failure)
- **Episode Day 7-9, One:**
 - **ACUTE, One:**
 - **Responder**, discharge expected today if clinically stable last 12h, **All:** ⁽³⁹⁾
 - **Anti-infective regimen, One:**
 - Complete
 - **Established and tolerated, One:**
 - Transition to PO anti-infective
 - Vascular access established for IV anti-infective
 - O₂ sat $\geq 92\%$ (0.92) or within acceptable limits ⁽⁴⁰⁾
 - Pulmonary function return to baseline or improving ^(7, 41)
 - Tolerating PO or nutritional route established ^(42, 43)
 - Treatment needs manageable in post acute setting ⁽⁴⁴⁾
 - **Complication or comorbidity, One:** ⁽⁴⁵⁾
 - No complication or active comorbidity relevant to this episode of care ⁽⁴⁵⁾
 - **Complication or comorbidity and clinically stable for discharge, $\geq$ One:**
 - **Acute kidney injury (AKI) and, $\geq$ One:**
 - Kidney function improving or within acceptable limits ⁽⁴⁰⁾
 - Outpatient dialysis regimen established
 - Adverse drug reaction resolving
 - Blood glucose within acceptable limits ⁽⁴⁰⁾
 - *C. difficile* infection, symptoms resolving, and tolerating PO ⁽⁶⁵⁾
 - Chest tube removed or Heimlich valve
 - Hemoptysis controlled or manageable
 - **Partial responder**, not clinically stable for discharge and requires continued stay, $\geq$ **One:**
 - **Inadequate clinical improvement and, Both:** ⁽⁴¹⁾
 - **Finding, $\geq$ One:**
 - FEV₁ $< 80\%$ (0.80) of baseline ^(9, 10)
 - Hemoptysis, persistent
 - O₂ sat $< 92\%$ (0.92) and $<$ baseline requiring supplemental oxygen ^(7, 19)
 - Pulmonary infiltrates worsening by imaging
 - Sputum production increasing
 - **Intervention, Both:**
 - **Airway clearance therapy, One:** ⁽²⁴⁾
 - At least 3x/24h
 - Contraindicated due to hemoptysis ⁽²⁵⁾
 - Anti-infective
 - **Anti-infective regimen incomplete and, Both:**

- Barrier to discharge, $\geq$ **One:** ⁽⁴⁴⁾
 - Patient or caregiver unable to adhere to treatment regimen
 - Treatment intensity exceeds resources provided in post acute setting
- Intervention, **Both:**
 - Airway clearance therapy, **One:** ⁽²⁴⁾
 - At least 3x/24h
 - Contraindicated due to hemoptysis ⁽²⁵⁾
 - Anti-infective
- AKI, **Both:** ⁽³⁰⁾
 - Finding, $\geq$ **One:**
 - Oliguria ^(46, 47)
 - Creatinine, **One:**
 - ≥ 1.5 x baseline and chronic kidney disease
 - ≥ 1.5 x ULN ⁽⁴⁸⁾
 - GFR $> 25\%$ (0.25) decrease from baseline ⁽⁷⁾
 - Dialysis initiated
 - Intervention, $\geq$ **One:**
 - Volume expander ≤ 2 d ^(49, 50)
 - Diuretic ≥ 2 x/24h, ≤ 3 d
 - Dialysis (initial) ≤ 7 d since initiation ^(51, 52)
 - Dialysis discontinued requiring lab monitoring ≤ 2 d since discontinuation ⁽⁵³⁾
 - Immunosuppressant therapy ≤ 3 d ^(54, 55, 56)
- Adverse anti-infective drug reaction requiring change in medication ≤ 24 h ⁽⁵⁷⁾
- Anti-infective level non-therapeutic and adjustment in dose or frequency ≤ 2 d
- Blood glucose ≤ 24 h, **All:** ⁽⁵⁸⁾
 - Finding, $\geq$ **One:**
 - > 250 mg/dL(13.9 mmol/L)
 - < 70 mg/dL(3.9 mmol/L)
 - Blood glucose monitoring
 - Insulin adjustment or dose held ^(58, 59)
- *C. difficile* infection, actual or suspected and, **Both:** ⁽⁶⁰⁾
 - Anti-infective (includes PO) ≤ 5 d ^(61, 62)
 - Inadequate oral intake ⁽⁶³⁾
- Chest tube and, $\geq$ **One:** ⁽⁶⁴⁾
 - Suction, continuous
 - Drainage ≥ 250 mL/24h
 - Repositioning within last 24h
 - Water seal within last 24h
- Continued weight loss and initiation or advancement of tube feeding ≤ 3 d
- Post Critical care ≤ 24 h
- INTERMEDIATE, $\geq$ **One:**
 - Anti-infective desensitization ≤ 24 h ⁽²⁹⁾
 - HFNC and respiratory monitoring every 3-4h ^(20, 21)
 - NIPPV ⁽²²⁾
 - Oxygen $\geq 40\%$ (0.40), ≤ 2 d ⁽¹⁹⁾
- CRITICAL, $\geq$ **One:**
 - AKI and, **Both:** ⁽³⁰⁾
 - Finding, $\geq$ **One:**
 - Potassium > 6.0 mEq/L(6.0 mmol/L) and, $\geq$ **One:**
 - Loss of P wave
 - Multifocal PVCs
 - Neuromuscular deficit ⁽³¹⁾
 - Widening QRS

- **Sodium, One:**
 - < 120 mEq/L(120 mmol/L) and volume overload
 - > 160 mEq/L(160 mmol/L)
- **Intervention, ≥ One:**
 - Calcium chloride
 - Calcium gluconate
 - Glucose 50%(0.50) with insulin ⁽³²⁾
 - Diuretic, continuous
- **Dialysis, One:**
 - Hemodialysis ≤ 7d since initiation
 - Peritoneal dialysis ≤ 7d since initiation
 - IV fluid resuscitation ≤ 2d ⁽³³⁾
- CRRT ⁽³⁴⁾
- DIC and blood product transfusion ⁽³⁵⁾
- ECMO or ECLS ⁽³⁶⁾
- **Hemodynamic instability unresponsive to ≥ 2 liter fluid bolus, requiring ongoing fluid resuscitation ≤ 24h and, ≥ One:** ⁽¹⁶⁾
 - MAP < 65 mmHg
 - Systolic blood pressure < 90 mmHg and < baseline
- Hemodynamic monitoring, invasive ⁽¹⁷⁾
- **Hemoptysis > 240ml/24h and, ≥ One:** ⁽¹⁸⁾
 - Lung resection required (**use General Surgical subset**) ⁽³⁸⁾
 - Blood product transfusion
 - Bronchial artery embolization performed within last 24h ⁽²⁸⁾
 - Post bronchial artery embolization monitoring ≤ 24h
- HFNC and respiratory monitoring every 1-2h ^(19, 20, 21)
- Mechanical ventilation
- **NIPPV and, ≥ One:** ⁽²²⁾
 - FiO₂ > 50%(0.50)
 - Respiratory intervention every 1-2h ⁽²⁷⁾
- Vasopressor, vasodilator, or inotrope, continuous (excludes low or fixed dose for end stage heart failure)
- **Episode Day 10, One:**
 - **ACUTE, One:**
 - **Responder, discharge expected today if clinically stable last 12h, All:** ⁽³⁹⁾
 - **Anti-infective regimen, One:**
 - Complete
 - **Established and tolerated, One:**
 - Transition to PO anti-infective
 - Vascular access established for IV anti-infective
 - O₂ sat ≥ 92%(0.92) or within acceptable limits ⁽⁴⁰⁾
 - Pulmonary function return to baseline or improving ^(7, 41)
 - Tolerating PO or nutritional route established ^(42, 43)
 - Treatment needs manageable in post acute setting ⁽⁴⁴⁾
 - **Complication or comorbidity, One:** ⁽⁴⁵⁾
 - No complication or active comorbidity relevant to this episode of care ⁽⁴⁵⁾
 - **Complication or comorbidity and clinically stable for discharge, ≥ One:**
 - **Acute kidney injury (AKI) and, ≥ One:**
 - Kidney function improving or within acceptable limits ⁽⁴⁰⁾
 - Outpatient dialysis regimen established
 - Adverse drug reaction resolving
 - Blood glucose within acceptable limits ⁽⁴⁰⁾
 - *C. difficile* infection, symptoms resolving, and tolerating PO ⁽⁶⁵⁾
 - Chest tube removed or Heimlich valve

- Hemoptysis controlled or manageable
 - **Non-responder**, not clinically stable for discharge and exceeds episode days at the Acute level of care for this condition, **One**:
 - **Use Extended Stay subset**
 - Refer for secondary review
 - **INTERMEDIATE, One**:
 - **Non-responder**, not clinically stable for discharge and exceeds episode days at the Intermediate level of care for this condition, **One**:
 - **Use Extended Stay subset**
 - Refer for secondary review
 - **CRITICAL, One**:
 - **Non-responder**, not clinically stable for discharge and exceeds episode days at the Critical level of care for this condition, **One**:
 - **Use Extended Stay subset**
 - Refer for secondary review
-

Discharge, Level of Care, ≥ One: ⁽⁶⁶⁾

- **HOME, Both**:
 - Level of care appropriateness, **Both**:
 - Home environment safe and accessible ⁽⁶⁷⁾
 - Patient or caregiver demonstrates ability to manage care
 - Cystic fibrosis discharge planning, **All**:
 - Provide comprehensive written discharge and teaching instructions ⁽⁶⁸⁾
 - Education on the importance of, **All**:
 - Recognizing the signs and symptoms of pulmonary exacerbation
 - Compliance with maintenance care
 - Infection prevention ⁽⁶⁹⁾
 - Medication reconciliation ⁽⁷⁰⁾
 - Follow-up care planned ⁽⁷¹⁾
 - Identify and address transportation needs
- **HOME CARE, Both**:
 - Level of care appropriateness, **All**:
 - Home environment safe and accessible ⁽⁶⁷⁾
 - Patient or caregiver willing and able to learn care
 - Outpatient management contraindicated or unavailable ^(72, 73)
 - Skilled services, ≥ **One**: ⁽⁷⁴⁾
 - Skilled nursing ⁽⁷⁵⁾
 - Skilled therapy, PT, OT, or ST
 - Behavioral health ⁽⁷⁶⁾
 - Cystic fibrosis discharge planning, **All**:
 - Provide comprehensive written discharge and teaching instructions ⁽⁶⁸⁾
 - Education on the importance of, **All**:
 - Recognizing the signs and symptoms of pulmonary exacerbation
 - Compliance with maintenance care
 - Infection prevention ⁽⁶⁹⁾
 - Medication reconciliation ⁽⁷⁰⁾
 - Follow-up care planned ⁽⁷¹⁾
 - Identify and address transportation needs
- **BEHAVIORAL HEALTH, Both**:
 - Level of care appropriateness, **One**:
 - Support systems available ⁽⁷⁷⁾
 - Skilled treatment, ≥ **One**:
 - Clinical assessment ⁽⁷⁸⁾

- Individual, group, or family counseling
- Psychiatric, substance, or medication evaluation

● **SKILLED MEDICAL OR THERAPY, Both:**

● Level of care appropriateness, **All:**

- Medical practitioner, NP, PA assessment or oversight $\geq 1\text{x/wk}$
- Treatment precluded in a lower level

● Services, $\geq$ **One:**

- Medical and skilled nursing interventions or assessment 1-2x/24h ⁽⁷⁵⁾

● Therapy, **All:**

- Goal directed therapy with rehabilitation potential ^(79, 80)
- Functional impairments requiring at least supervision ⁽⁸¹⁾
- Able to tolerate 1h/d of skilled therapy $\geq 5\text{d/wk}$

● Complete prior to facility transfer, **All:**

- Medication reconciliation ⁽⁷⁰⁾
- Obtain and complete forms for facility
- Obtain discharge summary and transmit to facility and medical practitioner
- Arrange transportation

● **SUBACUTE MEDICAL OR THERAPY, Both:**

● Level of care appropriateness, **All:**

- Medical practitioner, NP, PA assessment or oversight $\geq 2\text{x/wk}$
- Treatment precluded in a lower level

● Services, $\geq$ **One:**

- Medical and skilled nursing interventions or assessment 4h/24h ⁽⁷⁵⁾

● Therapy, **All:**

- Goal directed therapy with rehabilitation potential ^(79, 80)
- ≥ 2 functional impairments requiring at least minimum assistance ⁽⁸¹⁾
- Able to tolerate 2-3h/d of skilled therapy $\geq 5\text{d/wk}$ and ≥ 1 therapy discipline

● Complete prior to facility transfer, **All:**

- Medication reconciliation ⁽⁷⁰⁾
- Obtain and complete forms for facility
- Obtain discharge summary and transmit to facility and medical practitioner
- Arrange transportation

● **SUBACUTE REHAB, Both:**

● Level of care appropriateness, **All:**

- Medical practitioner, NP, PA assessment or oversight $\geq 3\text{x/wk}$
- Treatment precluded in a lower level

● Services, **All:**

- Goal directed therapy with rehabilitation potential ^(79, 80)
- ≥ 2 functional impairments requiring at least minimum assistance ⁽⁸¹⁾
- Able to tolerate 2-3h/d of skilled therapy $\geq 5\text{d/wk}$ and ≥ 2 therapy disciplines

● Complete prior to facility transfer, **All:**

- Medication reconciliation ⁽⁷⁰⁾
- Obtain and complete forms for facility
- Obtain discharge summary and transmit to facility and medical practitioner
- Arrange transportation

● **ACUTE REHAB, Both:**

● Level of care appropriateness, **Both:**

- Medical practitioner, NP, PA assessment or oversight $\geq 3\text{x/wk}$

● Services, **All:**

- Goal directed therapy with rehabilitation potential ^(79, 80)
- ≥ 2 functional impairments requiring at least minimum assistance ⁽⁸¹⁾
- Able to tolerate $\geq 15\text{h/wk}$ of skilled therapy and ≥ 2 therapy disciplines ⁽⁸²⁾

● Complete prior to facility transfer, **All:**

- Medication reconciliation ⁽⁷⁰⁾
- Obtain and complete forms for facility
- Obtain discharge summary and transmit to facility and medical practitioner
- Arrange transportation

● **LONG TERM ACUTE CARE, Both:**

- **Level of care appropriateness, Both:**
 - Treatment precluded in a lower level
- **In lieu of acute or continued hospitalization or failed lower level of care, ≥ One:**
 - Primary condition or illness and treatment of active comorbid condition(s) ⁽⁸³⁾
 - Ventilator dependent ≥ 6h/d and weaning planned ^(84, 85)
 - Acute and comorbid conditions requiring prolonged hospitalization ⁽⁸⁶⁾
- **Complete prior to facility transfer, All:**
 - Medication reconciliation ⁽⁷⁰⁾
 - Obtain and complete forms for facility
 - Obtain discharge summary and transmit to facility and medical practitioner
 - Arrange transportation

Utilization Benchmarks

Condition or Procedure	% Pd Obs	LOS (days)	Type
177 RESPIRATORY INFECTIONS AND INFLAMMATIONS WITH MCC	n/a	5	CMS GMLOS
178 RESPIRATORY INFECTIONS AND INFLAMMATIONS WITH CC	n/a	3.3	CMS GMLOS
179 RESPIRATORY INFECTIONS AND INFLAMMATIONS WITHOUT CC/MCC	n/a	2.5	CMS GMLOS
CYSTIC FIBROSIS	8%	8.2	InterQual

Percent Paid as Observation indicates the final payment status as a percentage for patients hospitalized with the condition. It may not correspond to the status assigned at admission. Note that patients initially placed in Observation status who are converted to inpatient status (common) are included in the inpatient category, and that patients initially admitted to inpatient status who are converted to observation status (uncommon) are included in the Observation category.

Notes:**1:**

Initial Review criteria are a level of care determination tool, intended to be used as **real-time** decision support in the emergency department to identify if observation or inpatient hospital level services are warranted. They help the reviewer determine whether a patient is appropriate for observation or inpatient admission at the time a decision to admit the patient is being made. Initial Review criteria evaluate **only data that are available at the time the decision is being made**. This may include previously provided interventions or the results of laboratory, imaging, and other tests. Initial Review may be appropriate when bridge or holding orders (e.g., admit to telemetry) are in place. These orders are intended to address the patient's needs until full treatment and medication orders are written.

Initial Review criteria **should not be used** retrospectively or once the decision to admit has been made (e.g., when full treatment and medication orders have been written). If the reviewer has sufficient information to complete an Episode Day 1 review, an Initial Review need not be completed.

Note: While Initial Review enables identification of the level of care, it is not intended to, and should not be used as a substitute for an Episode Day 1 review, which will generally include specific, evidence-based interventions and intensity of service requirements.

For further clarification, see the Acute Level of Care Review Process.

2:

Hemodynamically stable patients who require treatment, assessment, or intervention every 4 to 8 hours.

Note: InterQual® Level of Care criteria are intended to be used to support a patient-specific clinical determination of medical necessity and not for final billing or reimbursement decisions. Final payment decisions are based on a variety of factors including medical necessity, payer rules, and contracts.

3:

The precise definition of a pulmonary exacerbation remains uncertain, but a combination of clinical signs and symptoms can predict respiratory decline. Changes in cough, sputum production, work of breathing, and imaging can indicate a potential respiratory exacerbation (Plummer et al., Cochrane Database Syst Rev 2016, 9: Cd006682).

4:

Expected treatments for a pulmonary exacerbation of cystic fibrosis include: increasing the frequency of airway clearance, treatment with inhaled bronchodilators, chest physical therapy and postural drainage, use of antibiotics, and increasing the dose of mucolytic agents (Plummer et al., Cochrane Database Syst Rev 2016, 9: Cd006682).

5:

Failed treatment is defined as, at least 1 week on a new anti-infective regimen accompanied by an increased frequency or dose of nebulizer, without evidence of clinical improvement.

6:

Hemoptysis is a common symptom in patients with cystic fibrosis. It can occur during an acute pulmonary exacerbation (most commonly with *P. aeruginosa* positive sputum culture) and can range from scant to life-threatening amounts. In most cases, hemoptysis will resolve spontaneously or during treatment of the respiratory exacerbation. Interventions to control bleeding, such as bronchial artery embolization, are reserved for unstable patients with massive hemoptysis (West et al., J Cyst Fibros 2017:).

7:

Baseline refers to either the patient's normal baseline or a newly established baseline. In the absence of documentation, a patient's baseline status may be presumed to be normal.

8:

Vital signs are considered sustained findings when they are abnormal for 2 or more readings greater than 15 minutes apart or are lasting at least 15 minutes. This excludes an isolated reading, a transient abnormal measurement, or a finding that requires urgent treatment (e.g., severe hypoxemia, ventricular arrhythmia, hypotension). Application of criteria for a vital sign finding should be based on confirmatory measurements

repeated at regular intervals. Events that are infrequent or vital sign abnormalities that have resolved with outpatient treatment are not considered to be sustained (e.g., resolution of hypertension following anti-hypertensive therapy, resolution of tachycardia following rehydration).

Note: The definition of sustained reflects the opinion of InterQual's® external peer reviewers. This criteria point is based upon current best practice and is the product of an iterative process involving multiple clinicians with diverse expertise in varied geographic settings.

9:

Baseline forced expiratory volume over one second (FEV₁) is the standard method used to establish the progression of disease in a patient with cystic fibrosis. The baseline FEV₁ refers to the patient's documented value (within 3 months) prior to the development of an acute pulmonary exacerbation (Plummer et al., Cochrane Database Syst Rev 2016, 9: Cd006682).

10:

In an effort to address inequities in medicine and improve appropriate access to care, multiple medical societies have been reevaluating the use of adjustments made for a patient's race and updating their calculators and reference equations when reporting diagnostic results.

In 2023, the American Thoracic Society issued updated guidance on reporting of pulmonary function tests (PFTs), advocating for a race-neutral average reference equation to determine the baseline (i.e., predicted normal) for a given patient (Bhakta et al., Am J Respir Crit Care Med 2023, 207: 978-995). The reference equation previously included lower predicted normal values for patients of certain races and ethnicities (in the US, for Black and Asian patients); a practice which has not been found to have biological validity and may lead to misclassification of diagnoses and disease severity in racial or ethnic minorities (Vyas et al., N Engl J Med 2020, 383: 874-882). Since not all spirometer manufacturers and PFT laboratories have switched over to the new race-neutral average reference equation, clinical judgement and consideration of patient-specific factors should be used when interpreting PFT results and deciding whether patients with borderline reported PFT values in the presence of concerning symptoms or signs should have further evaluation or treatment.

11:

The Centers for Disease Control and Prevention (CDC) define a fever as greater than or equal to 100.4 degrees Fahrenheit (i.e., greater than or equal to 38 degrees Celsius) and do not specify the route of measurement (Centers for Disease Control and Prevention, Definitions of symptoms for reportable illness. 2017). The route utilized may be directed by institutional protocols, equipment availability, patient preference, or other patient-specific factors.

12:

During a pulmonary exacerbation, patients should receive a nutrition consult to ensure optimization of nutrition, which is essential to meeting increased energy demands and promoting recovery. Nutrition management may include consuming an unrestricted diet high in calories and protein, pancreatic enzyme replacement therapy, and supplementation. Patients with cystic fibrosis-related diabetes should follow the same dietary guidelines described above. In order to achieve glycemic control, carbohydrate counting and use of insulin-to-carbohydrate ratios are recommended (Moran et al., Diabetes Care 2010, 33: 2697-708).

13:

Hemodynamically stable patients who require treatment, assessment, or intervention every 2 to 4 hours.

Note: InterQual® Level of Care criteria are intended to be used to support a patient-specific clinical determination of medical necessity and not for final billing or reimbursement decisions. Final payment decisions are based on a variety of factors including medical necessity, payer rules, and contracts.

14:

Mental status change may include confusion, disorientation, delirium, or increasing lethargy (Lacey et al., Ann Med 2019, 51: 232-51).

Instruction: Patients with acute coma, stupor, or obtundation should be reviewed at a higher level of care due to the need for more frequent neurological monitoring. These criteria exclude chronic coma, stupor, or obtundation (Shenvi et al., Ann Emerg Med 2020, 75: 136-45).

15:

Hemodynamically unstable patients (or those with the potential to become unstable) who require treatment, assessment, or intervention every 1 to 2 hours.

Note: InterQual® Level of Care criteria are intended to be used to support a patient-specific clinical determination of medical necessity and not for final billing or reimbursement decisions. Final payment decisions are based on a variety of factors including medical necessity, payer rules, and contracts.

16:

Fluid resuscitation for patients with sepsis-induced hypoperfusion or hypotension usually begins with 30 mL per kg in adults; the recommendation for this starting amount was downgraded from a strong to a weak recommendation in the 2021 Surviving Sepsis Campaign Guidelines as a change from the 2016 Guidelines (Evans et al., Crit Care Med 2021, 49: e1063-e143; Rhodes et al., Crit Care Med 2017, 45: 486-552; Semler and Rice, Clin Chest Med 2016, 37: 241-50). Traditionally, parameters used to assess response to fluid administration included improvements in mean arterial pressure (MAP), urine output, or central or mixed venous oxygen saturation. Surviving Sepsis Campaign guidelines recommend that decisions regarding additional fluid administration should be based on frequent reassessment of hemodynamic and clinical status, as well as dynamic variables of fluid responsiveness including inferior vena cava diameter, pulse pressure variation, stroke volume variation, and lactate clearance (Evans et al., Crit Care Med 2021, 49: e1063-e143; Rhodes et al., Crit Care Med 2017, 45: 486-552).

Patients with ongoing hypotension and inability to tolerate the recommended amount of fluid due to volume overload may need early initiation of vasopressor therapy.

17:

Invasive hemodynamic monitoring may include at least one of the following methods of assessment: arterial line, pulmonary artery catheter, central venous catheter, or cardiac output monitoring (Rajaram et al., Cochrane Database Syst Rev 2013, 2: CD003408).

18:

Massive hemoptysis in Cystic Fibrosis is defined as, a quantity of greater than 240 ml over 24 hours, blood loss that is life threatening, or repeated large amounts of bleeding over a short period of time (days) (Hurt and Simmonds, Paediatr Respir Rev 2012, 13: 200-5).

19:

Oxygen therapy is the administration of oxygen at concentrations greater than ambient air (room air: 21%(0.21)) with the intent of treating and/or preventing the symptoms and manifestations of hypoxia. The oxygen concentration or percentage (FiO₂) delivered varies with the manufacturer's design, oxygen flow rate, the patient's respiratory rate, and tidal volume. Actual inspired FiO₂ with nasal cannula varies significantly with the patient's minute ventilation and pattern of breathing. For patients at risk for CO₂ retention, where a precise inspired FiO₂ is required, the Venturi mask is the preferred method. In general, patients who are very ill or have respiratory disease may require considerably higher flow rates to achieve the desired FiO₂. The following are estimates of O₂ delivered at the associated flow rates:

LOW-FLOW SYSTEMS**Nasal cannula**

Room air 21%(0.21)

1L 24%(0.24) **4L** 36%(0.36)

2L 28%(0.28) **5L** 40%(0.40)

3L 32%(0.32) **6L** 44%(0.44)

Simple oxygen masks

- 35-50%(0.35-0.50) FiO₂ (5-10 L/min)
- Flow rates are usually maintained at 5 L/min or more to avoid accumulation of CO₂ in the mask

Venturi masks

- 24-50%(0.24-0.50) FiO₂ (4-12 L/min)

Partial rebreathing masks

- 40-70%(0.40-0.70) FiO₂ (6-10 L/min)

Non-rebreathing masks

- 60-80%(0.60-0.80) FiO₂ (minimum flow of 10 L/min)

HIGH-FLOW SYSTEMS**Air-entrainment masks/nebulizers**

- 24-40%(0.24-0.40) FiO₂

Heated, humidified high-flow concentrating nasal cannula (HFNC)

- 24-100%(0.24-1.0) FiO₂

20:

High-flow nasal cannula (HFNC) delivers a fraction of inspired oxygen (FiO₂) that is heated (up to 37 degrees Celsius) with 100%(1.0) relative humidity at flow rates up to 60 liters per minute, set independent from the FiO₂. The heated and humidified gas has been shown to improve comfort and aid in clearance of secretions. The high-flow rates reduce anatomical dead space by flushing out carbon dioxide from the upper airways, leading to improved work of breathing. In addition, HFNC reduces the amount of ambient air a patient can breathe in, resulting in less oxygen dilution. A Cochrane review was conducted in 2021 to determine the effectiveness of HFNC compared to standard oxygen therapy and noninvasive positive pressure ventilation (NIPPV) in adult patients requiring respiratory support in the intensive care unit (ICU). The systematic review found that HFNC had less treatment failure compared to standard oxygen therapy (i.e., face mask or nasal cannula with a flow rate of 15 L/min); however, it did not influence mortality rate or length of stay. When HFNC was compared to NIPPV, which included bilevel positive airway pressure (BiPAP) or continuous positive airway pressure (CPAP), there was no difference in treatment failure defined as intubation or re-intubation, mortality, or length of stay. In conclusion, HFNC has been shown to have less treatment failure compared to standard oxygen therapy and may be an alternative to NIPPV in patients treated for acute hypoxic respiratory failure (Lewis et al., Cochrane Database Syst Rev 2021, 3: Cd010172).

21:

Respiratory monitoring includes an evaluation of the patient's respiratory rate, heart rate, oxygen saturation, or work of breathing. Within 90 minutes of high-flow nasal cannula (HFNC) initiation, the patient's respiratory status should improve, indicating adequate respiratory support is being provided. If clinical improvement is not achieved within this time frame, additional respiratory support may be indicated (Lee et al., Intensive Care Med 2013, 39: 247-57).

22:

Noninvasive positive pressure ventilation (NIPPV), also known as NIV, provides respiratory support by application of a tightly fitting facial mask, nasal mask, or helmet rather than an endotracheal tube or tracheostomy. In some cases, the use of NIPPV can avoid the need for endotracheal intubation and decreases the risk of barotrauma, lung injury, and/or infection. NIPPV is commonly delivered by a bilevel positive airway pressure ventilator (BiPAP), a continuous positive airway pressure device (CPAP), or a mechanical ventilator. Supplemental oxygen can be delivered at concentrations approximating 100%(1.0).

The decision to provide respiratory support via NIPPV, and the modality to provide NIPPV, is based upon the patient's specific clinical findings (e.g., medical condition leading to respiratory failure, underlying comorbidities, clinical progression, improvement). Examples of NIPPV devices are volumetric (i.e., deliver a defined volume), barometric (i.e., deliver a defined pressure), and combined (i.e., deliver defined volumes and pressure). The terminology for NIPPV delivery systems may differ between equipment manufacturers and provider organizations. InterQual® criteria do not differentiate between the different NIPPV modalities.

Whether or not high-flow nasal cannula (HFNC) should be classified as a component of NIPPV is unclear, and there is conflicting evidence in the medical literature. Where HFNC is an appropriate modality, InterQual® defines this in a specific criteria point. As such, NIPPV does not include HFNC (Hackett, A., PulmCCM 2018; Nardi et al., F1000Res 2017, 6: 290; Osadnik et al., Cochrane Database Syst Rev 2017, 7: CD004104; Allison and Winters, Emerg Med Clin North Am 2016, 34: 51-62; Gregoretti et al., Crit Care Clin 2015, 31: 435-57).

23:

Guidelines recommend continuing maintenance medication regimens for cystic fibrosis patients in the treatment of pulmonary exacerbations. Nebulizer treatments are a major component of maintenance therapy and may include aerosolized antibiotics, mucolytics, and bronchodilators. Practitioners must carefully consider the risk of drug interactions when deciding to continue a maintenance medication that may adversely interact with a newly

initiated medication (Mogayzel and Flume, Am J Respir Crit Care Med 2010, 181: 539-44).

24:

Airway clearance therapy (ACT) is used to optimize pulmonary function by clearing sputum from the respiratory tract. The most effective techniques are:

- Chest physiotherapy (CPT): Percussion and vibration of the chest and can be combined with modified postural drainage (body positioning without the head down) to remove mucus from the lungs.
- Positive expiratory pressure (PEP): A mask or mouthpiece and breathing against applied pressure (10-20 cm H₂O). Oscillating PEP (not recommended in children), vibrates the airflow during PEP
- Active-cycle-of-breathing technique (ABCT): A cycle of deep controlled breathing, and forced thoracic expiration to clear secretions (used in all stages of disease)
- Autogenic drainage (AD): A method of controlled breathing to move mucus from the small airways to the larger airways for expulsion
- High-frequency chest compression (use of a compressive vest to vibrate the airways)

There is insufficient evidence to recommend one technique over another, and treatment should be individualized according to age, patient preference, and risk for adverse effects. The Cystic Fibrosis Foundation recommends that airway clearance therapy increase in duration or frequency during the treatment of pulmonary exacerbations (Strickland et al., Respir Care 2015, 60: 1071-7).

25:

Contraindicated refers to **documented** clinical circumstances when a medication or treatment is not indicated based on the nature of the disease, comorbid condition, or a drug allergy or intolerance.

26:

Instruction: Room air assessment may not be clinically appropriate in patients who are dyspneic, tachypneic, cyanotic, or experiencing moderate to severe respiratory distress. If a patient remains hypoxic while on oxygen, room air assessment is not required.

27:

Respiratory interventions include ventilator/noninvasive positive pressure ventilation (NIPPV) setting changes, chest physiotherapy, suctioning, and nebulizer or metered-dose inhaler (MDI) treatments.

28:

Bronchial artery embolization should be performed in cystic fibrosis patients with massive hemoptysis who are clinically unstable. The aim of this treatment is to embolize the bleeding vessel in an effort to stop the bleeding (Flume et al., American journal of respiratory and critical care medicine 2010, 182: 298-306).

29:

Desensitization is the safest measure of reintroducing a drug when alternative medicines are not tolerated. Due to frequent anti-infective exposure, cystic fibrosis patients often develop allergic reactions to multiple agents. Desensitization protocols involve the administration of increasing concentrations and doses of an anti-infective medication in frequent intervals. Patients are monitored closely for anaphylaxis and bronchospasm (Whitaker et al., J Cyst Fibros 2011, 10: 282-5; Legere et al., Journal of cystic fibrosis: official journal of the European Cystic Fibrosis Society 2009, 8: 418-24).

30:

Acute kidney injury (AKI) is a term used to describe the broad spectrum of kidney function impairment, including insufficiency, oliguria, and failure. The spectrum ranges from minor changes in renal function markers (e.g., increase in serum creatinine, decreased urine output) to failure requiring renal replacement therapy.

AKI is defined as a sudden decrease in renal function resulting from multiple etiologies, including intrinsic kidney disease, ischemia, nephrotoxicity, and extrarenal pathology.

AKI encompasses patients with chronic kidney disease who experience an acute deterioration. AKI is associated with significant morbidity and mortality in hospitalized patients (Ostermann et al., Kidney Int 2020, 98: 294-309; Guideline Updates, In: Acute kidney injury: prevention, detection and management. 2019; Kidney Disease:

Improving Global Outcomes (KDIGO), Kidney International Supplements 2012, 2: ii.-138; Wang et al., Am J Nephrol 2012, 35: 349-55).

31:

Hyperkalemia may be associated with weakness leading to muscle twitching, paresthesia, and paralysis.

32:

Dextrose 50%(0.50) with insulin is used for the treatment of hyperkalemia. This treatment shifts potassium intracellularly, and repeated doses can be given if the hyperkalemia persists. Other treatments that may be used simultaneously include potassium binding agents, diuretics, nebulized albuterol, calcium, sodium bicarbonate, and dialysis (Depret et al., Ann Intensive Care 2019, 9: 32).

33:

Fluid resuscitation involves the administration of repeated intravenous fluid boluses to acutely restore depleted intravascular volume. There is no evidence to support the use of colloids over crystalloids; studies have demonstrated that the primary choice for a resuscitation fluid is a balanced crystalloid solution, such as Lactated Ringer's (Myburgh, N Engl J Med 2018, 378: 862-3).

Bolus volumes for administration may be generally defined by the following:

- Fluid boluses of 250 to 1,000 milliliters in adults (Scales, Br J Nurs 2014, 23; Myburgh and Mythen, N Engl J Med 2013, 369: 1243-51; Vincent and De Backer, N Engl J Med 2013, 369: 1726-34)
- Fluid boluses of 10 to 20 milliliters per kilogram in pediatric patients. In some cases, boluses of 60 milliliters per hour or more may have to be administered in the first hour (Davis et al., Crit Care Med 2017, 45: 1061-93)

For patients with decreased cardiac reserve, significant renal dysfunction, or third-spacing, smaller volume boluses may be used (National Clinical Guideline Centre for Acute and Chronic Conditions, IV Fluids in Children:

Intravenous Fluid Therapy in Children and Young People in Hospital. 2015. Reaffirmed 2020). The total number of fluid boluses required to restore intravascular volume and avoid fluid overload varies by patient (Joseph et al., J Trauma Acute Care Surg 2014, 76: 457-61; Holodinsky et al., Crit Care 2013, 17: R249; Kirkpatrick et al., Intensive Care Med 2013, 39: 1190-206). Dynamic measures such as bedside vena cava ultrasound and monitoring of urine output may assist in determining when volume resuscitation has been achieved (Kent et al., J Surg Res 2013, 184: 561-6; Zengin et al., Am J Emerg Med 2013, 31: 763-7; Weekes et al., Acad Emerg Med 2011, 18: 912-21).

34:

While no randomized, controlled trials have shown that continuous renal replacement therapy (CRRT) is superior to intermittent dialysis with respect to survival, there are several advantages that may influence its use despite the lack of demonstrated survival benefit.

These advantages include:

- Gradual solute and fluid shifts resulting in greater hemodynamic stability, little risk of disequilibrium syndrome, and no worsening of brain edema
- Less need for fluid or nutritional restrictions, allowing for adequate nutrition without compromising fluid balance
- Decreased risk for intradialytic hypotension and recurrent kidney injury

Comprehensive guidelines recommend the use of CRRT for patients with hemodynamic instability, brain edema, persistent metabolic acidosis, and large fluid removal requirements. CRRT can be discontinued once renal recovery has been confirmed or another form of renal replacement is chosen based on the patient's clinical condition (Ostermann et al., Kidney Int 2020, 98: 294-309; Depret et al., Ann Intensive Care 2019, 9: 32; Kidney Disease: Improving Global Outcomes (KDIGO), Kidney International Supplements 2012, 2: ii.-138).

35:

Disseminated intravascular coagulation or DIC is a hematological emergency where there is abrupt systemic cascade activation of coagulation leading to thrombosis of midsize to small blood vessels rapidly leading to acute organ dysfunction. Simultaneously, there is consumption of platelets and proteins involved in the coagulation cascade, resulting in acute thrombocytopenia and low clotting factors. DIC can be caused by conditions such as sepsis, trauma, severe obstetrical complications, and malignancies. Laboratory abnormalities found in DIC include

low fibrinogen (not always present), elevated fibrin degradation products, elevated D-dimer, thrombocytopenia, and prolonged prothrombin (PT) and activated partial thromboplastin times. Prompt therapy for DIC includes treatment against the underlying disease but may also include platelet transfusion, plasma (fresh frozen plasma; FFP) or factor transfusion in cases of hemorrhage, or anticoagulants in situations with thrombosis formation predominates (Wada et al., J Thromb Haemost 2013:).

36:

Extracorporeal membrane oxygenation (ECMO) or extracorporeal life support (ECLS) provides oxygen and removes carbon dioxide from blood. ECMO/ECLS facilitates the recovery of cardiopulmonary function when other measures fail to improve gas exchange. Indications for the use of ECMO/ECLS in children include refractory septic or cardiogenic shock, cardiac arrest as a bridge to transplant, or refractory hypoxemic or hypercarbic respiratory failure. There is clear evidence to support the use of ECMO/ECLS in the pediatric population, but definitive studies in adults are lacking. However, in adults, observational studies and other data support the use of ECMO/ECLS for severe refractory, hypoxemic, or hypercarbic respiratory failure (Mehta et al., American College of Cardiology Perspectives and Analysis 2018; Napp et al., Herz 2017, 42: 27-44; Guirand et al., J Trauma Acute Care Surg 2014, 76: 1275-81). ECMO/ECLS incorporates the use of an external circuit, a cannula for vascular access, a pump to propel the blood through the circuit, and an artificial membrane that oxygenates the blood. Common complications include hemorrhage, hemolysis, and infection. Stroke, embolism, and seizures in children have also been reported. Infants with intraventricular hemorrhage (IVH) or those at high risk for IVH (i.e., weight less than 2000 grams, age less than 35 weeks) are not candidates for this therapy (Abrams et al., J Am Coll Cardiol 2014, 63: 2769-78; Domico et al., Pediatr Crit Care Med 2012, 13: 16-21; MacLaren et al., Intensive Care Med 2012, 38: 210-20; Peek et al., Health Technol Assess 2010, 14: 1-46; Mugford et al., Cochrane Database Syst Rev 2008: CD001340).

37:

Invasive hemodynamic monitoring may include at least one of the following methods of assessment: arterial line, pulmonary artery catheter, central venous pressure, or cardiac output monitoring (Rajaram et al., Cochrane Database Syst Rev 2013, 2: CD003408).

38:

In the treatment of cystic fibrosis patients with massive hemoptysis, lung resection should be performed as a last resort after all other measures have failed (e.g., bronchial artery embolization) (Flume et al., American journal of respiratory and critical care medicine 2010, 182: 298-306).

39:

Selection of this criteria point indicates that the patient is responding to treatment and is clinically stable for transfer or discharge. To determine the most appropriate post-acute level of care, see discharge criteria.

40:

When a criteria point states "within acceptable limits," it refers to either the patient's normal baseline, a newly established baseline, or parameters that the medical practitioner determines are acceptable.

41:

Patients admitted with pulmonary exacerbations may not return to prior respiratory function with treatment. At risk patients for permanent pulmonary decline include those with advanced disease and multiple prior exacerbation episodes (Cogen et al., Pediatrics 2017, 139:).

42:

Routes for nutrition may be by mouth (PO), intravenously (IV), jejunostomy tube (J-tube), or gastrostomy tube (G-tube).

43:

Oral intake should be at least equivalent to the patient's maintenance fluid requirements or within parameters which the medical practitioner deems appropriate. Maintenance fluid requirements should be increased in the presence of dehydration or insensible loss. Maintenance fluid rates are based on ideal body weight and reflect the

minimum amount of fluid required to maintain hydration. Rates may be decreased in older patients or those with a history of renal impairment or heart failure (Scales, Br J Nurs 2014, 23).

44:

Since treatment of a pulmonary exacerbation involves a combination of aggressive respiratory care and intravenous anti-infective therapy, continued hospitalization is recommended if the resources and support required to treat the exacerbation exceed that which can be provided in the post acute setting (Cogen et al., Pediatrics 2017, 139:). Barriers to discharge may include the following:

- Patient and/or caregiver skill or compliance with the treatment regimen in question
- Medication administration and airway clearance therapy cannot be maintained in the post acute setting at the prescribed frequency or intensity

45:

Relevant complications or comorbidities are conditions that require active management during an inpatient stay.

46:

In this instance, oliguria is not the result of an outlet obstruction.

47:

Urine output less than 0.5 mL/kg/h may be considered oliguria.

48:

This line of criteria refers to a patient without chronic kidney disease and an unknown creatinine baseline.

49:

Volume expanders are fluids administered intravenously to increase circulatory volume. Studies have demonstrated that a balanced crystalloid solution (e.g., Lactated Ringer's) is preferable to colloids in restoration of intravascular volume in sepsis and septic shock (Evans et al., Crit Care Med 2021, 49: e1063-e143; Rhodes et al., Crit Care Med 2017, 45: 486-552; Winters et al., J Emerg Med 2017, 53: 928-39). However, crystalloids have been found to be less effective than colloids at stabilizing hemodynamic endpoints in critically ill patients (Martin and Bassett, J Crit Care 2019, 50: 144-54). The type of fluid selected for administration should be based on the indication for its use and other patient-specific factors.

Instruction: In order to apply criteria for volume expanders, there should be documentation of a volume deficit supported by clinical findings. The volume of infusion is patient-specific and varies based on the cause of volume depletion, comorbid condition, and patient response. Criteria for volume expander should not be applied for maintenance intravenous fluids or electrolyte replacement.

50:

For expansion of intravascular volume in patients with acute kidney injury (AKI), guidelines recommend initial treatment with an isotonic crystalloid solution (sodium chloride or sodium bicarbonate); the effects of sodium chloride on renal outcomes are being studied. Forced diuresis is not recommended because it can worsen AKI in volume-responsive patients (Ostermann et al., Kidney Int 2020, 98: 294-309; Guideline Updates, In: Acute kidney injury: prevention, detection and management. 2019; Pistolesi et al., J Nephrol 2018, 31: 797-812; Joannidis et al., Intensive Care Med 2017, 43: 730-49; Kidney Disease: Improving Global Outcomes (KDIGO), Kidney International Supplements 2012, 2: ii.-138).

51:

Initial course refers to the initiation of a chronic hemodialysis/peritoneal dialysis program or initiation for acute renal pathology. A repeat initiation for a recurrence or exacerbation of an acute renal problem, provided that there was a gap of more than 14 days between treatments, would also qualify as an initial course.

52:

Dialysis includes hemodialysis or peritoneal dialysis. Both modalities are equally effective and should be provided

based on patient needs and device availability (Liu et al., Cochrane Database Syst Rev 2017, 12: Cd011457).

53:

Following the discontinuation of dialysis, lab monitoring is necessary for continued management and may include blood urea nitrogen (BUN), creatinine, and electrolytes.

54:

Immunosuppressant medications are used to treat conditions such as graft-versus-host disease (GVHD), hemolytic anemia, organ transplant, graft failure, rejection, inflammatory bowel disease, acute glomerulonephritis, and inflammatory cellulitis. Medications used for immunosuppression include prednisone, prednisolone, cyclosporine, azathioprine, mycophenolate mofetil, tacrolimus, sirolimus, everolimus, belatacept, antithymocyte globulin (ATG), and basiliximab.

55:

Instruction: If a patient is clinically stable, has no other medical reason for an acute level of care, and has tolerated 24 to 48 hours of PO immunosuppressants (after conversion from IV), consider a less intense level of care setting.

56:

Prior to the initiation of immunosuppressants for systemic diseases, which adversely impacts kidney function, a renal biopsy may be necessary.

57:

Anti-infective medications may cause adverse drug reactions. Symptoms of an adverse drug reaction include urticaria, angioedema, bronchospasm, pruritus, rash, vomiting, diarrhea, anaphylaxis, and thrombocytopenia.

58:

Guidelines recommend preprandial glucose targets of 80-130 mg/dL (4.4-7.2 mmol/L) and a random blood glucose target range of 100-180 mg/dL (5.6-10.0 mmol/L) in hospitalized patients. In stable patients with previous tight glycemic control, more stringent targets may be appropriate. In patients with severe comorbidities, a history of hypoglycemia, terminal illness, late onset diabetes, and in older patients, less stringent targets may be appropriate. The preferred method for achieving and maintaining glucose control in non-critically ill patients is scheduled subcutaneous insulin administration. Guidelines discourage the sole use of sliding scale insulin in the inpatient hospital setting. Rather, an insulin regimen consisting of basal, prandial, and correctional insulin is recommended in hospitalized patients (American Diabetes Association, Diabetes Care 2024, 47: S1-S314).

59:

Instruction: Insulin adjustments should be based on blood glucose values obtained by lab draw or glucose monitor. Intermittent insulin must be administered at least 3 times in 24 hours. If the patient is on a continuous insulin infusion, the rate must be adjusted at least twice in 24 hours. A dose that is held based on a normal or low blood glucose value should be included in the count of doses administered within the 24-hour period.

60:

Clostridioides difficile is the most common cause of nosocomial infectious diarrhea, particularly in patients receiving anti-infectives (Johnson et al., Clin Infect Dis 2021, 73: e1029-e44).

61:

Clostridioides difficile is the most common cause of nosocomial infectious diarrhea, particularly in patients receiving anti-infectives. Guidelines describe multiple acceptable regimens but do indicate a preference for oral fidaxomicin for patients with mild to moderate disease. Alternative regimens are driven by disease severity and the number of episodes the patient has experienced. Treatment guidelines recommend discontinuation of the inciting anti-infective as soon as possible to prevent recurrence. Other interventions include IV fluids, electrolytes, or oral fluids to maintain hydration. Use of opiates and antidiarrheal medications should be avoided because decreased intestinal motility will exacerbate the infection (Johnson et al., Clin Infect Dis 2021, 73: e1029-e44).

62:

Instruction: This criteria point is intended for anti-infectives initiated for the treatment of *C. difficile* infections. The preferred treatment of *Clostridioides difficile* infections is based on the episode of illness and the severity of the symptoms. The Infectious Diseases Society of America (IDSA) recommends a variety of treatment regimens which are dependent on clinical presentation, including disease severity and how many episodes of disease the patient has experienced (Johnson et al., Clin Infect Dis 2021, 73: e1029-e44).

63:

Inadequate oral intake refers to the inability to maintain hydration according to a patient's documented fluid goal or maintenance fluid requirements. The inability to maintain hydration may occur when excessive output (e.g., vomiting, diarrhea, or insensible losses) leads to a net fluid deficit (Scales, Br J Nurs 2014, 23).

64:

Chest tubes are used to evacuate air, fluid, or blood from the pleural space. For malignancies involving the pleura, chest tubes may also be used for the administration of sclerosing agents. Serial chest x-rays may be performed to ensure that there is no re-accumulation of air or fluid. The chest tube may be removed when there is no evidence of air leak, output has slowed to 250-400 mL/day, and the lung is fully expanded (Vuong et al., Respir Med 2018, 137: 152-66).

65:

Signs indicating recovery from *Clostridioides difficile* include:

- Decrease in the severity and frequency of diarrhea
- Absence of fever
- Normal white blood cell count

Not all patients demonstrate fever or leukocytosis during the course of illness. Diet advancement should be based on improving clinical signs and symptoms.

66:

The discharge screen is a resource tool and not criteria. Referring to the discharge screen at the initiation of discharge planning is recommended.

67:

The home environment and safety assessment will normally include an evaluation of the patient's pre- and post-hospitalization functional level, physical layout of the home (e.g., entrances and exits, stairs, access to the community), identification of unsafe conditions (e.g., scatter rugs, missing handrails, oxygen use and smoking, lack of fire safety devices, inadequate lighting, heating, and cooling), factors that may trigger symptoms (e.g., secondhand smoke, poor food choices, dysfunctional family dynamics), sanitation hazards (e.g., lack of electricity, running water, refrigeration, inadequate toilet facilities, presence of insects or rodents), and the need for adaptive equipment (e.g., walker, handrails for the tub or shower, elevated toilet seat).

68:

Ensuring the patient and/or caregiver understands all aspects of the condition and can assume responsibility for self-care is crucial in preventing readmission. Assessing a patient and/or caregiver's level of understanding after education has been provided can be difficult. The teach-back method is an effective way of providing education at the appropriate level and can also be used to assess the learner's comprehension. To use the teach-back method, discuss key points in common terms and avoid using medical jargon or unfamiliar terms. After the education has been delivered, ask the learner to repeat what was learned. Gaps or misinterpretations in the learner's explanation will pinpoint areas where communication may have failed and provide opportunity for clarification.

69:

In order to avoid subsequent pulmonary exacerbations, prevention techniques include vaccination, avoidance of crowds, and frequent hand washing.

70:

Medication reconciliation is a formal process or technique used by health care providers and pharmacists to identify the most complete and accurate list of all medications a patient is taking at times of transitions in care (e.g., upon hospital admission, transfer from one unit to another during hospitalization, or discharge from the hospital to home or another facility). The goals of this process are to ensure medication and dosages are appropriate for the patient, resolve discrepancies in drug regimens, and ultimately prevent medication errors and reduce adverse drug events. Medication reconciliation is a Joint Commission National Patient Safety goal. Coordinating information when a patient is transferred to another setting, service, practitioner, or level of care ensures accurate medications are listed. The process is comprised of the following (Hospital: National Patient Safety Goals. 2020):

- Obtain an external list of medications (e.g., medications taken prior to admission)
- Develop a list of current medications and add them to the medical record
- Compare the medications from the external list to the current list
- Clarify inconsistencies/discrepancies (e.g., omissions, duplications, contraindications, unclear information, and changes)
- Develop a list of medications to be prescribed at discharge or transfer
- Communicate the new list to the patient and/or appropriate caregiver(s)
- Ensure that patient and/or caregiver(s) understand the medication information upon discharge

Specific medication issues identified as being problematic include missing medication information from transfer orders, lack of information on medications provided in the acute setting, incomplete medication records, discrepancies between hospital regimen and discharge summary, and missing information pertaining to the patient's tolerance of a medication regimen. Although medication reconciliation is an important aspect in patient safety, there is a lack of consensus and evidence about the best effective methods of implementing this process. Physician-led and electronic medication reconciliation in hospitals are effective strategies to reduce medication discrepancies, however, the impact of these interventions is uncertain due to the low quality of evidence (Choi and Kim, J Clin Pharm Ther 2019, 44: 932-45; Redmond et al., Cochrane Database Syst Rev 2018, 8: CD010791). Trained pharmacy technicians, under the direction of a licensed pharmacist, may be an option for developing and expanding medication reconciliation processes (Irwin et al., Hosp Pharm 2017, 52: 44-53).

71:

Close follow-up of older adults, pediatric, or at-risk patients after discharge can help minimize hospital readmission and total health care costs (Rising et al., Ann Emerg Med 2013, 62: 145-50; Frei-Jones et al., Pediatr Blood Cancer 2009; 52(4): 481-485; Kripalani et al., JAMA 2007; 297(8): 831-841). Studies have shown that for greater than 50% (0.50) of patients re-hospitalized within 30 days of discharge, there was no visit to a physician's office between the time of discharge and re-hospitalization (Hernandez et al., JAMA 2010, 303: 1716-22; Jencks et al., N Engl J Med 2009; 360(14): 1418-1428). Efforts to reduce readmissions should focus on patients known to be at high risk and patients who have frequent visits to the emergency department (ED). Patients at high risk may have multiple ED visits prior to readmission (Rising et al., Ann Emerg Med 2013, 62: 145-50).

72:

Outpatient management contraindicated refers to the following:

- The patient whose medical condition is so fragile that going to the clinic or medical practitioner's office for treatment would put the patient at risk
- The patient with extensive equipment (e.g., ventilators, specialized beds) where travel outside the home would be cumbersome and places the patient at risk
- The treatment goal is to adapt the home environment for increased independence (e.g., an individual who is blind who needs mobility education)

73:

Outpatient management unavailable refers to the following:

- The patient whose treatment (e.g., dressing changes) is so extensive that it takes too long or is too frequent (e.g., twice daily) to be done in an outpatient clinic or medical practitioner's office
- The scheduling of the treatment falls outside the medical practitioner's office or clinic hours
- Travel distance to receive care is excessive (e.g., greater than 1 hour)
- The service is unavailable in the outpatient setting

74:

Skilled nursing services refer to services that must be provided by a licensed nurse qualified to assess and monitor patient condition(s) and provide medical treatment and/or teaching to patients who have skilled nursing needs. Examples of Home Care level of care interventions include skilled assessment, intervention, or education for any of the following:

- New treatment or medication regimen
- Adjustment in the treatment or medication regimen
- Infusion therapy administration, teaching, or access device management
- Management and evaluation of a care plan for patients with an active comorbidity and multiple care needs
- Chronic disease requiring a disease management program (e.g., asthma, heart failure, chronic obstructive pulmonary disease, diabetes)
- End-stage disease, hospice, or palliative care

75:

Skilled nursing interventions refer to services that are provided or supervised by a qualified or licensed professional when the care needs of the patient require assessment, observation, monitoring, and/or education to achieve the medically desired result. Close observation and assessment by nursing may be necessary at this level of care. This includes nursing care such as medication administration (excludes routine), vital signs monitoring, and/or other interventions not listed.

76:

The Home Care criteria are used for a patient who has been admitted or is expected to be admitted for home care. When criteria are met for this level of care and it is not available in your area, we recommend you use the next higher level of care.

Home care can include specialty services provided by a psychiatric nurse following a specialized treatment plan developed by a psychiatrist or advanced practice registered nurse. It is intended for treatment and monitoring of a patient's psychiatric condition that prevents the patient from leaving the house or makes leaving the house extremely difficult or unsafe (Centers for Medicare & Medicaid Services. Chapter 7, In: Medicare benefit policy manual. 2022). Home care can also be used for a patient with severe and persistent mental illness with a history of nonadherence to medication.

Evaluation and Treatment

Programming may differ based upon legislative and geographical variances and is subject to organizational policy; however, at a minimum, it should include:

- Care coordination with treating physician and other providers
- Clinical assessment every visit
- Education every visit
- Medication assessment, teaching, and management every visit
- Medication reconciliation initiated within first visit
- Psychosocial assessment within first visit
- Substance evaluation within first 2 visits

For those patients who are covered under a Medicare benefit, the Affordable Care Act mandates that a certifying physician or allowed practitioner have a face-to-face visit with the patient within 90 days prior to the start of care, or within 30 days after the start of care. There must be documentation of the patient's condition that supports the need for home care services and the requirement of homebound status (Centers for Medicare & Medicaid Services. Chapter 7, In: Medicare benefit policy manual. 2022).

77:

Support system includes friends, family, social services, health care delivery providers or agencies, sponsors, clergy, and self-help groups.

78:

The initial clinical assessment is performed by a licensed professional and includes a comprehensive review of the patient's presenting diagnosis and a review of body systems. Identification of current and potential medical needs and health problems can be identified by the clinical assessment. The clinical assessment includes:

- History and physical exam (e.g., vital signs, height, weight)
- Pain assessment includes cause, intensity, quality, onset, duration, and effects on quality of life (e.g., activities of daily living (ADLs), instrumental activities of daily living (IADLs), and interpersonal relationships). Pain relievers should also be included
- Nutritional and hydration status
- Functional ability
- Safety and infection control measures
- Prescribed and over-the-counter (OTC) medications, herbal supplements, and home remedies
- Patient's and/or caregiver's understanding of medication use, dosing, side effects, and adherence to the medication or treatment regimen
- Patient's and/or caregiver's understanding of the illness or disease process and potential long-term complications
- Patient's and/or caregiver's coping strategies to deal with the illness or injury and ability to follow the plan of care

79:

Goal-directed therapy includes one or more of the following:

- Activities of daily living (ADLs) (e.g., feeding, dressing, bathing, grooming, toileting, functional mobility)
- Bed mobility or transfers (e.g., transfers to chair, toilet, commode, tub, shower, car)
- Cognitive, speech, language, or swallowing retraining
- Home and lifestyle modification (e.g., assessment and training focused on the home, job, school, or work environment)
- Pain management techniques (e.g., assessment and interventions used to relieve pain or decrease its intensity)
- Pulmonary rehabilitation (e.g., energy conservation, speech, breathing re-education, postural drainage, percussion, or vibration)
- Positioning and splinting
- Range of motion and strengthening
- Wheelchair mobility
- Ambulation

80:

Rehabilitation potential refers to the probability that therapy and medical goals are realistic and attainable based on patient's prior level of function, severity of illness or injury, and the extent of impairments.

81:

Functional assistance levels are based upon the patient's function during tasks and activities necessary to return to household mobility or ambulation. The functional assistance level required for each individual task or activity (e.g., mobility, activities of daily living (ADLs)) may vary.

The following terms are commonly used in the post-acute setting:

- Independent - Patient can safely and within a reasonable amount of time perform a task (or developmentally appropriate task) without physical or cognitive assistance or supervision
- Modified Independent - Patient performs an activity with a supportive device, adaptive equipment, and/or prosthetic or orthotic device. Additional time may be required to complete the activity and/or there are safety (risk) considerations
- Supervision - Patient performs an activity with standby or distant supervision or setup. Verbal cueing or coaxing, without physical contact or setup of items and application of orthoses may be required when patient's safety awareness is impaired
- Minimum or limited Assistance - Patient performs at least 75%(0.75) of an activity and requires some physical contact to steady, guide, or move
- Moderate or extensive Assistance - Patient performs at least 50%(0.50) of an activity and requires physical assistance for functional mobility or ADLs
- Maximum Assistance - Patient performs 25%(0.25) to 50%(0.50) of an activity and requires physical assistance for functional mobility or ADLs
- Total Assistance or dependence - Patient performs less than 25%(0.25) of an activity and may require total assistance for functional mobility or ADLs

82:

The intensity of rehabilitation therapies is a driver for admission to an acute inpatient rehabilitation facility. Documentation should reflect the need for required and ongoing intensive therapies. The determination of whether a patient requires and can tolerate an intensive rehabilitation program of at least 15 hours is not based on any one single piece of documentation. All clinical documentation should be evaluated by the reviewer to determine whether the patient is demonstrating the need for services and has the ability to tolerate them. Patients who do not require this level of intensive therapy should be treated at a lower level of care. Rehabilitation therapies should be reasonable and medically appropriate and should not exceed a patient's level of tolerance. If there are clinical features which preclude the patient's ability to tolerate the intensity threshold of at least 15 hours per week, secondary review would be appropriate (Centers for Medicare & Medicaid Services, Medicare Benefit Policy Manual Chapter 1 - Inpatient Hospital Services Covered Under Part A. 2021; Miller and Forer, Pm r 2021, 13: 535-9; Forrest et al., Medicine (Baltimore) 2019, 98: e17096).

83:

Comorbid conditions include new onset or worsening behavioral symptoms, heart failure, deep vein thrombosis, uncontrolled diabetes, immunocompromised states, malignant or end-stage disease, malnutrition, renal insufficiency or end-stage renal disease, infection, ventilator dependence, noninvasive positive pressure ventilation or respiratory insufficiency, complex wound care, and new functional impairment(s) with limitation.

84:

Ventilation includes invasive ventilation and noninvasive positive pressure ventilation (NIPPV), including continuous positive airway pressure (CPAP) and bilevel positive airway pressure (BiPAP).

85:

Notable causes for prolonged mechanical ventilation include:

- Ventilator-acquired pneumonia
- Renal failure
- Heart failure
- Exacerbation of chronic obstructive pulmonary disease (COPD)
- Chest wall disorder
- Neuromuscular diseases
- Traumatic brain injury
- Sepsis
- Complications following cardiac surgical procedures

Major contributing factors for failure to wean include (Ambrosino and Vitacca, Multidiscip Respir Med 2018, 13: 6; Davies et al., Br J Anaesth 2017, 118: 563-9):

- Increased work of breathing
- Impaired respiratory drive
- Inspiratory muscle weakness

86:

Long-Term Acute Care (LTAC) is a recognized level of care designation by the Centers for Medicare and Medicaid Services (CMS) for acute care hospitals. It is defined by federal statutes as a level of care for patients requiring an average length of stay of 25 days or more (Centers for and Medicaid Services, Fed Regist 2018, 83: 41144-784). Patients who require a short stay for treatment may be more appropriate for the subacute or skilled nursing facility (SNF) level of care.